Bean diseases

Fungal diseases

1. Cercospora leaf spot - *Cercospora cruenta*
2. Rust - *Uromyces phaseoli typica, U. appendiculatus*
3. Powdery mildew - *Erysiphe polygoni*
4. Anthracnose - *Colletotrichum lindemuthianum*
5. Stem and pod anthracnose *Colletotrichum truncatum*
6. Angular leaf spot - *Phaeoisariopsis griseola*
7. Root rot and web blight- *Rhizoctonia solani (PS:Thanatephorus cucumeris)*
8. Charcoal rot or dry root rot- *Macrophomina phaseolina*

Bean diseases

Bacterial diseases

1. Common bacterial blight/ Fuscus blight -*Xanthomonas axanopodis* pv. *phaseoli*.
2. Halo blight / Greasy spot - *Pseudomonas savastanoi* pv. *phaseolicola*
3. Bacterial wilt - *Curtobacterium flaccumfaciens*

Viral diseases

1. Bean common mosaic - *Bean Common Mosaic Virus* (BCMV)
2. Bean yellow mosaic-*Bean Yellow Mosaic Virus*, *Mungbean yellow virus* or *Phaseolus virus*-2 (ss DNA virus)

Cercospora leaf spot - *Cercospora cruenta*

Cercospora leaf spot - *Cercospora cruenta*

Symptoms

- Usually on older leaves with circular to angular, brown spots on lamina.
- The centre of the spot is grey with reddish border or dark brown.
- Several spots coalesce to form bigger spots.
- The central portion of the spot dries and falls off giving **shot hole** appearance.
- **Rarely pods** are also affected.

Cercospora leaf spot - Cercospora cruenta

Disease cycle and Management

Refer to earlier notes of Cercospora .

Rust - *Uromyces phaseoli typica, U. appendiculatus*

Rust - *Uromyces phaseoli typica, U. appendiculatus*

Symptoms:

- The symptoms **mostly appear on leaves**, though **often found on petiole, stem and pods** of some hosts
- On leaves, the disease may appear on both the surfaces, but is more common on the abaxial surface (lower surface).
- Pustules of the disease usually appear first as small and slightly raised spots that are almost white in colour.
- These pustules enlarge by forming **reddish brown sori,** up to 2 mm in diameter, containing the urediniospores.

Rust - *Uromyces phaseoli typica, U. appendiculatus*

Symptoms contd…

- After **coalescing,** they may occupy larger areas.
- **A ring of secondary sori** may develop around the original infection on susceptible cultivars.
- Later in the season with the formation of teliospores, the sori turn **dark brown or black**.
- The affected leaves may turn yellow and dry or fall prematurely.

Rust - *Uromyces phaseoli typica, U. appendiculatus*

Pathogen:

- Being autoecious rust, all its spore stages are produced on bean plant but **pycnial and aecial stages are not commonly encountered in nature**.
- The urediniospores are **brownish, globoid or ellipsoid, one celled and echinulate.**
- Spore wall is **golden brown** 1-1.5 µm thick having two equatorial pores.

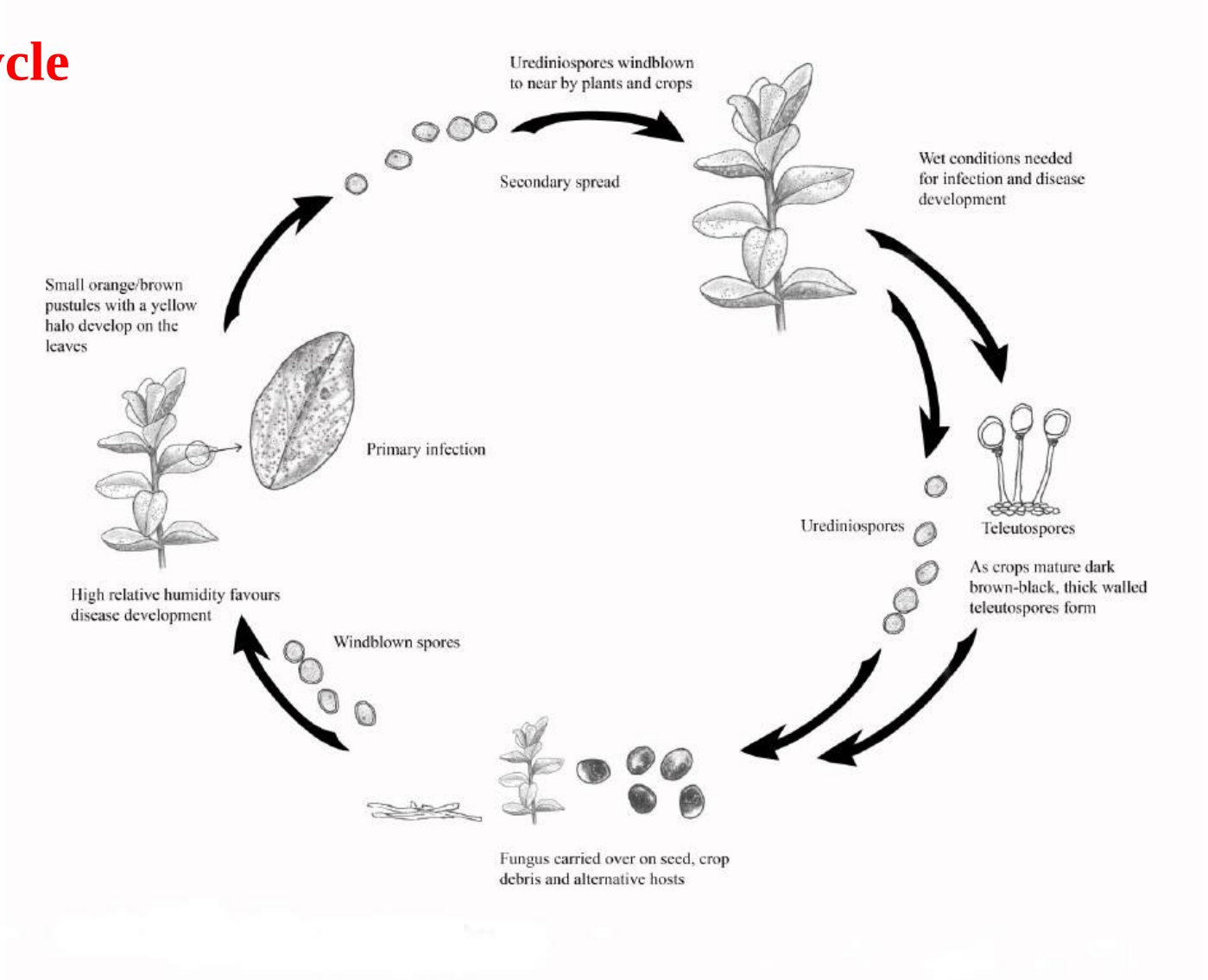
Rust - *Uromyces phaseoli typica, U. appendiculatus*

Pathogen contd

- Teliospores are one celled, **pedicellate**, globoid or broadly ellipsoid.
- They may be smooth or possessing a few **verucose** marks on the wall.
- The teliospores wall is **chestnut brown** in colour with **a hyaline papilla** over the pore.

Rust - *Uromyces phaseoli typica, U. appendiculatus*

Life cycle



Rust - *Uromyces phaseoli typica, U. appendiculatus*

Disease cycle and epidemiology:

- In areas where the crop is grown continuously and collateral hosts are also present, inoculum in the form of urediniospores is available round the year.
- The life cycle of the fungus is completed on the host itself; however, in cooler areas where bean is taken as single crop in the year, teliospores surviving in crop debris serve as source of perpetuation.
- Areas having high humidity and temperature upto 24°C are most suitable for disease development.
- **Urediniospores germinate** optimally at **15-24° C** while for **teliospores germination 10-15°C** is optimum.

Rust - *Uromyces phaseoli typica, U. appendiculatus*

Management:

- Collect and destroy the infected plant debris and collateral hosts
- Adjust sowing dates to avoid severe infection of plants in the field
- Follow long crop rotations.
- With the initiation of the disease, spray the crop with mancozeb (0.25%) or hexaconazole (0.1%) or difenoconazole (0.05%) or combination of mancozeb (0.25%) and hexaconazole (0.05%) and repeat at 10 days interval.

Powdery mildew - *Erysiphe polygoni*

Powdery mildew - *Erysiphe polygoni*

Symptoms:

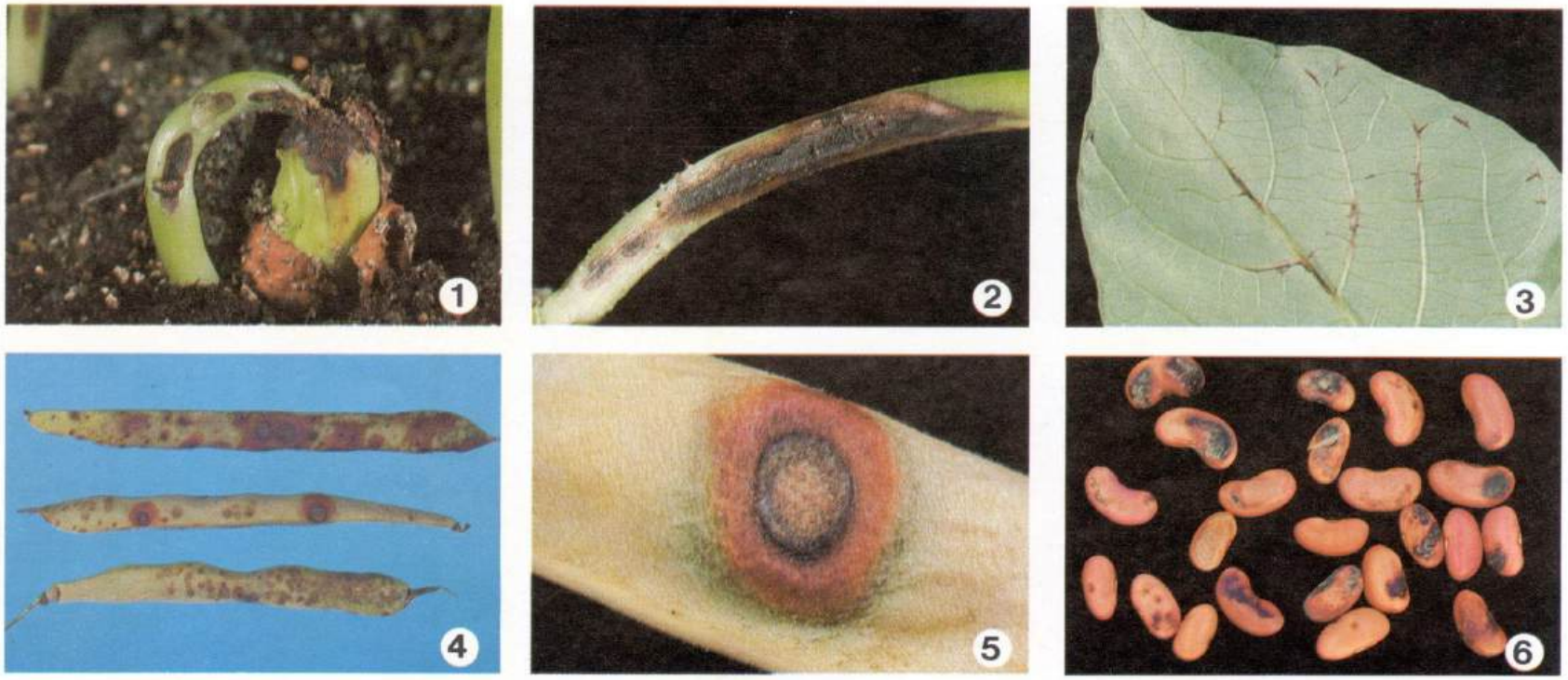
- Powdery patches appear on both surfaces of leaves.
- These patches coalesce and the leaf blade turns brown and dry up.
- Under severe conditions young pods and stems are also covered with powdery coating.
- Affected young pods turn black and shrivel.
- Seeds are not filled properly.

Powdery mildew - *Erysiphe polygoni*

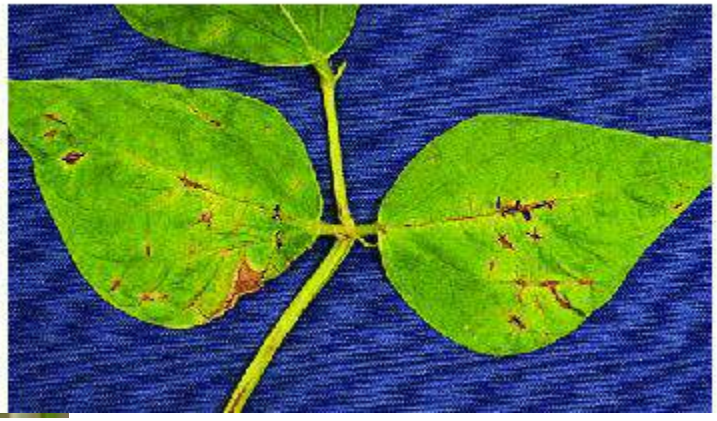
Disease cycle and Management

Refer earlier PM fungi

Anthracnose - *Colletotrichum lindemuthianum*



Anthracnose - *Colletotrichum lindemuthianum*



Anthracnose - *Colletotrichum lindemuthianum*

Anthracnose - *Colletotrichum lindemuthianum*

a. Seeds b. Leaves c. green pods d. Anthracnose-infected plants in the field
e. symptoms of anthracnose on mature pods

Anthracnose - *Colletotrichum lindemuthianum*

- The disease is more severe in temperate and subtropical mountaneous regions of the world with cool and wet climate.

Symptoms:

- All the above ground parts are affected at any stage of crop growth. However, the characteristic symptoms appear on pods
- On cotyledons spots are sunken dark brown or black with pink spore mass
- Seedling infection results in **collapse of seedling**

Anthracnose - *Colletotrichum lindemuthianum*

Symptoms contd..

On leaves

- On leaves, the infection appear on the **under leaf surface as blackened dead portions of the veins** which may extend to limited adjoining areas.

- Later these may also appear on upper surface. When the infection is severe, the affected plants wither off.

Anthracnose - *Colletotrichum lindemuthianum*

Symptoms contd...

On immature pods

- Most striking symptoms as brown and sunken **circular spots of varying sizes with bright red, yellow or orange margins**
- The centre of the spots later **turns grey or pink** due to sporulation of the pathogen especially in wet weather.
- The spots **with raised border**
- In severely infected pods, brown to light chocolate coloured **sunken cankers on seed coat**.

Stem and pod anthracnose - *Colletotrichum truncatum*

Symptoms:

- Whole plant looks chlorotic, stunted
- On stem reddish elongated cankers girdle the stem and cause death of the part above infection
- Cankers appear along the veins first on the lower surface of the leaves
- On pods- reddish small streaks or blotches
- Spots on pods later turn pale brown to grayish and thickly covered with black fruiting bodies

Anthracnose - *Colletotrichum lindemuthianum*

Pathogen:

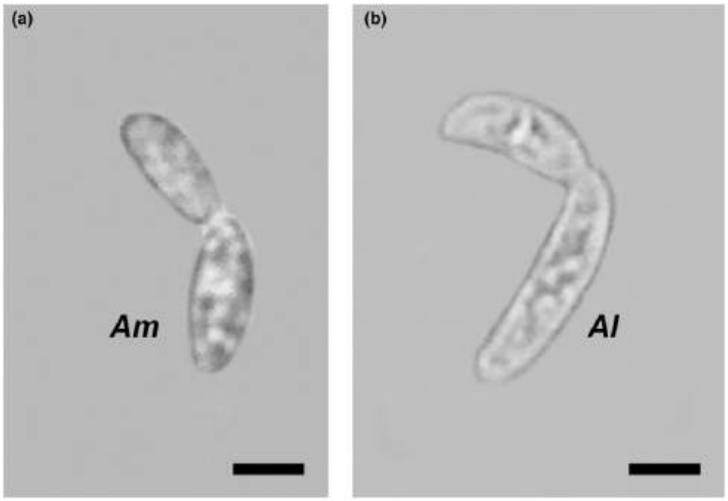
- The mycelium is branched, septate and hyaline at first becoming dark with age.
- The acervulus contains a layer of 3 to 50 conidiophores depending on the size of lesion.
- Conidia are borne successively and acrogenously on short conidiophores under favourable conditions until pinkish spore masses appear on the surface and setae are produced sparingly.

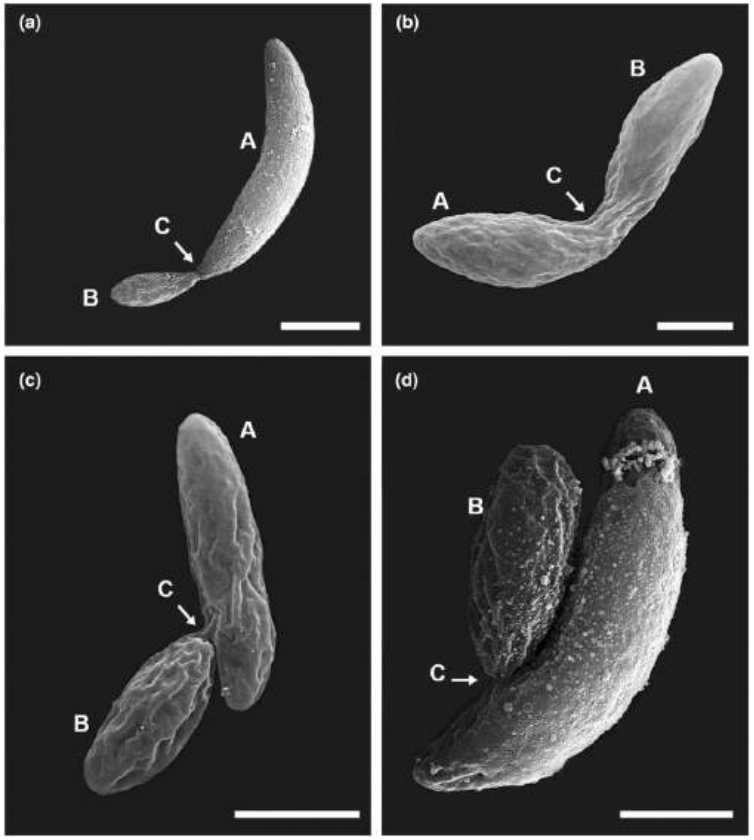
Anthracnose - *Colletotrichum lindemuthianum*

Pathogen contd…

- Conidia are hyaline, cylindrical having rounded ends or somewhat pointed at one end and often bear a clear vacuole like body near the centre.
- The teliomorph of the pathogen has been identified as *Glomerella cingulata* (=*G. lindemuthianum*).
- Perithecia contain hyaline and filiform periphyses and asci.
- Each ascus contains eight **allantoid ascospores**, which are ejected from the tip of ascus.

G. lindemuthianum





Budding of allantoid ascospores

Anthracnose - *Colletotrichum lindemuthianum*

Disease cycle and epidemiology:

- The pathogen perpetuate in infected seeds and plant debris
- When infected seed germinates, lesions appearing on cotyledons which serve as the source of secondary inoculum producing the spores as water borne.
- **Primary leaves and the hypocotyl** are foci of secondary infections.
- The teliomorph of the fungus is rarely found in nature.
- It develops most abundantly in cool, wet weather and largely disappears under hot and dry conditions.

Anthracnose - *Colletotrichum lindemuthianum*

Disease cycle and epidemiology contd…

- A relative humidity of 92 % and above is essential for infection, the optimum being close to 100%.
- **The fungus requires about 10 mm of rain to establish initial infection.**
- The optimum temperature for disease development ranges from **18 to 27°C with maximum intensity at 21°C** and is markedly reduced at 13°C.
- The movement of insects, animals and man may spread conidia particularly when foliage is moist.

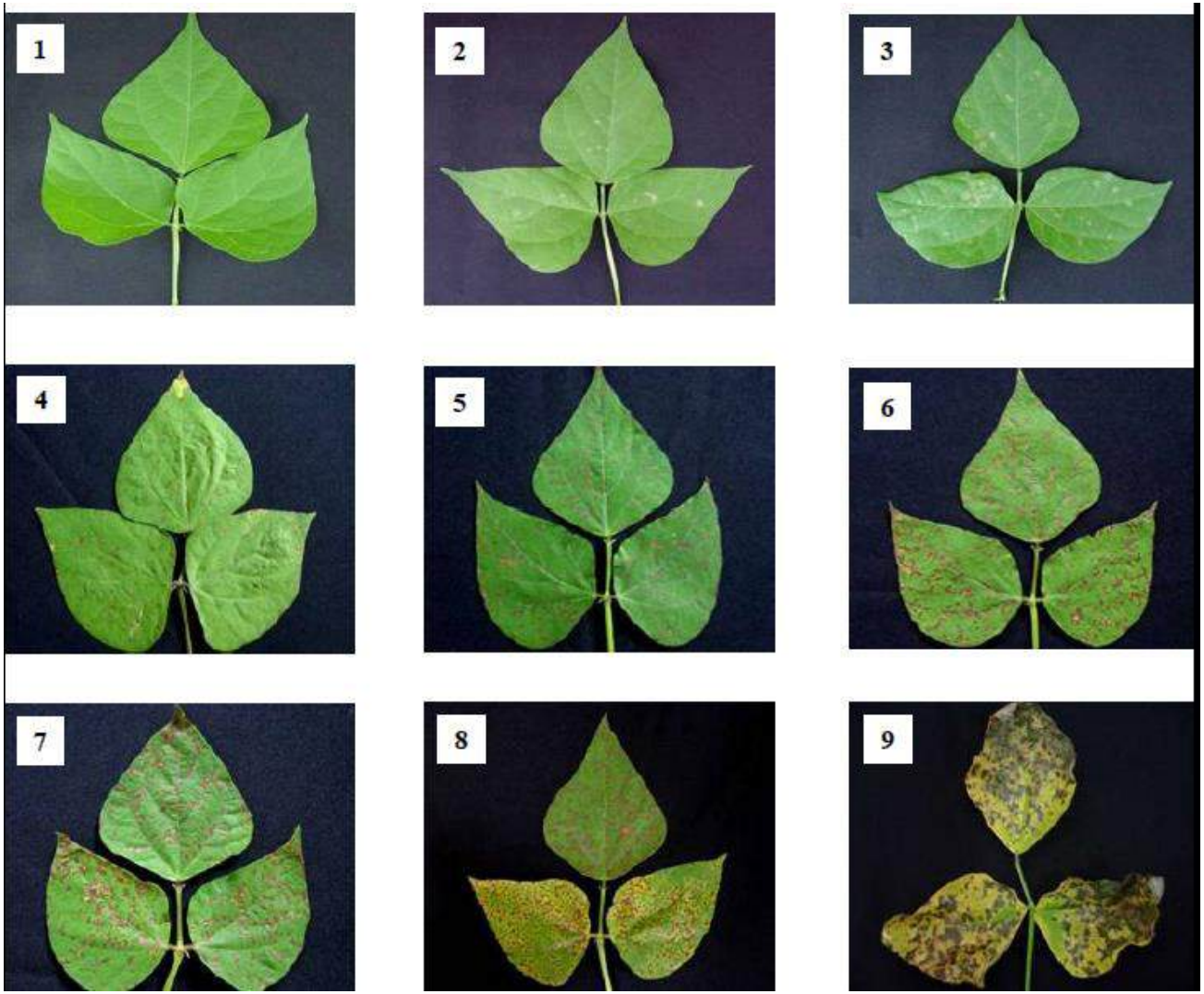
Anthracnose - *Colletotrichum lindemuthianum*

Management:

- Follow 2-3 years of crop rotation with non-leguminous crops.
- Use disease free seed and treatment with carbendazim or benomyl (0.2%).
- Resistant cultivars/ lines like KRC 1, KRC 17, EC 42960 and EC 57080
- In field, the spread of the disease can be limited by application of fungicides like mancozeb (0.25%) or combination of mancozeb (0.25%) and carbendazim (0.05%) or tricyclazole (0.03%) and repeat at 10 -14 days interval.

Angular leaf spot - *Phaeoisariopsis griseola*

Angular leaf spot - *Phaeoisariopsis griseola*



Angular leaf spot - *Phaeoisariopsis griseola*

Angular leaf spot - *Phaeoisariopsis griseola*

- The disease was first reported by Saccardo (1878) from Italy.
- In India disease was first noticed from Nilgiri hills by Srinivasan, whereas from H.P. by Sohi *et al*. (1963).

Symptoms: Leaves, petioles, stems, pods and seed

- The symptoms appear as circular spots on cotyledonary leaves
- On true leaves 3-5 angled, dark grayish coloured spots on the upper surface appear in between veins and veinlets while light gray on the lower surface.

Angular leaf spot - *Phaeoisariopsis griseola*

Symptoms contd…

- Later on the spots change to reddish brown and finally to dark brown colour and on close observation, in the spots coremia bearing large number of spores are present.
- In severe infections, **leaves show upward curling** and defoliate prematurely
- **Elongated dark brown lesions also appear on stem and petioles.**

Angular leaf spot - *Phaeoisariopsis griseola*

Symptoms contd…

- The fungus also attacks the pods causing superficial, smooth, usually circular spots with **reddish brown center and ashy black borders** but with **time deeper tissues are also involved.**

- **Infection from pods also goes to the developing seed.**

- Underneath the pod lesions **yellowish brown discolouration on infected seeds.**

Angular leaf spot - *Phaeoisariopsis griseola*

Pathogen:

- *Phaeoisariopsis griseola* (Syn: *Isariopsis griseola*).
- **Sexual stage of the fungus has not yet observed.**
- The fungus produces a stromatic structure in the substomatal cavity.
- Columnar hyphae producing conidiophores are laterally aggregated into parallel bundles in the form of coremia or synnemata which are darker at the base and lighter towards the apex and either curved, flexuous or septate

Angular leaf spot - *Phaeoisariopsis griseola*

C. Fasciculate conidiophores.
D. Conidia.

Angular leaf spot - *Phaeoisariopsis griseola*

Pathogen contd…

- Conidia are light grey to hyaline, cylindrical to spindle shaped and straight to slightly curved, 1-3 (rarely 4-6) septate which are borne singly on the tips of conidiophore.

Disease cycle and epidemiology:

- The pathogen perpetuate both on seed and in infected plant debris
- Infected seed germinates to give rise to infected plantlets bearing coremia of the fungus on the undersurface of the cotyledonary leaves.
- The conidia released from the coremia disseminate either by rain splashes or with wind blown soil particles.

Angular leaf spot - *Phaeoisariopsis griseola*

Disease cycle and epidemiology contd…

- The conidia germinate to form germ-tube, which enters through stomata and advances intercellularly in the mesophyll and palisade parenchyma.
- **Within 7-10 days of infection, the fungus produces necrotic lesions**, develops stromata in the substomatal cavity, which may sporulate during 24 to 48 h of continuous moisture.
- The conidia continue the cycle by acting as secondary inoculum.

Angular leaf spot - *Phaeoisariopsis griseola*

Disease cycle and epidemiology contd…

- Optimum temperature for growth and sporulation, spore germination and development of the disease are 21-24°C, 18-24°C and 24°C, respectively while, optimum RH for sporulation, spore germination and symptoms development is >90 per cent.

Angular leaf spot - *Phaeoisariopsis griseola*

Management:

- Collection and destruction of crop debris.
- Crop rotation with non-leguminous crops at least 2-3 years
- Use of healthy seed and treatment with carbendazim (0.2%).
- With the initiation of the disease, spray the crop with mancozeb (0.25%) or combination of mancozeb (0.25%) and carbendazim/ benomyl/ thiophanate methyl (0.05%) or hexaconazole (0.05%) and repeat at 10-14 days interval.

Root rot and web blight- *Rhizoctonia solani* (PS:*Thanatephorus cucumeris*)

Rhizoctonia root rot **causes rotting of lateral roots**, leaving the taproot and secondary roots

Root rot and web blight- *Rhizoctonia solani* (PS:*Thanatephorus cucumeris*)

Web blight affects leaves and pods. Symptoms on leaves initially water soaked, followed by **browning or necrotic circular lesions**. The infected leaves die and fall off.

Root rot and web blight- *Rhizoctonia solani* (PS:*Thanatephorus cucumeris*)

Root rot and web blight- *Rhizoctonia solani* (PS:*Thanatephorus cucumeris*)

Pod symptoms are **brown coloured irregular lesions**.
The lesions may merge to form **large spots.**
The seeds in affected pods will be damaged.

Phaeoisariopsis griseola

Rhizoctonia solani

Colletotrichum lindemuthianum

Root rot and web blight- *Rhizoctonia solani* (PS:*Thanatephorus cucumeris*)

- Disease is of economic importance especially when crop is grown **during rainy season and is more prevalent on bush type than pole type bean.**

Symptoms: Disease appear in following two phases:

Root rot:

- Symptoms appear on **roots and stem** above and below the surface of the soil as reddish brown, **sunken cankers**

- The **lesions enlarge rapidly and girdle the stem** at the collar region, extend downward to the roots which rots partially or completely

Root rot and web blight- *Rhizoctonia solani* (PS:*Thanatephorus cucumeris*)

Web blight:

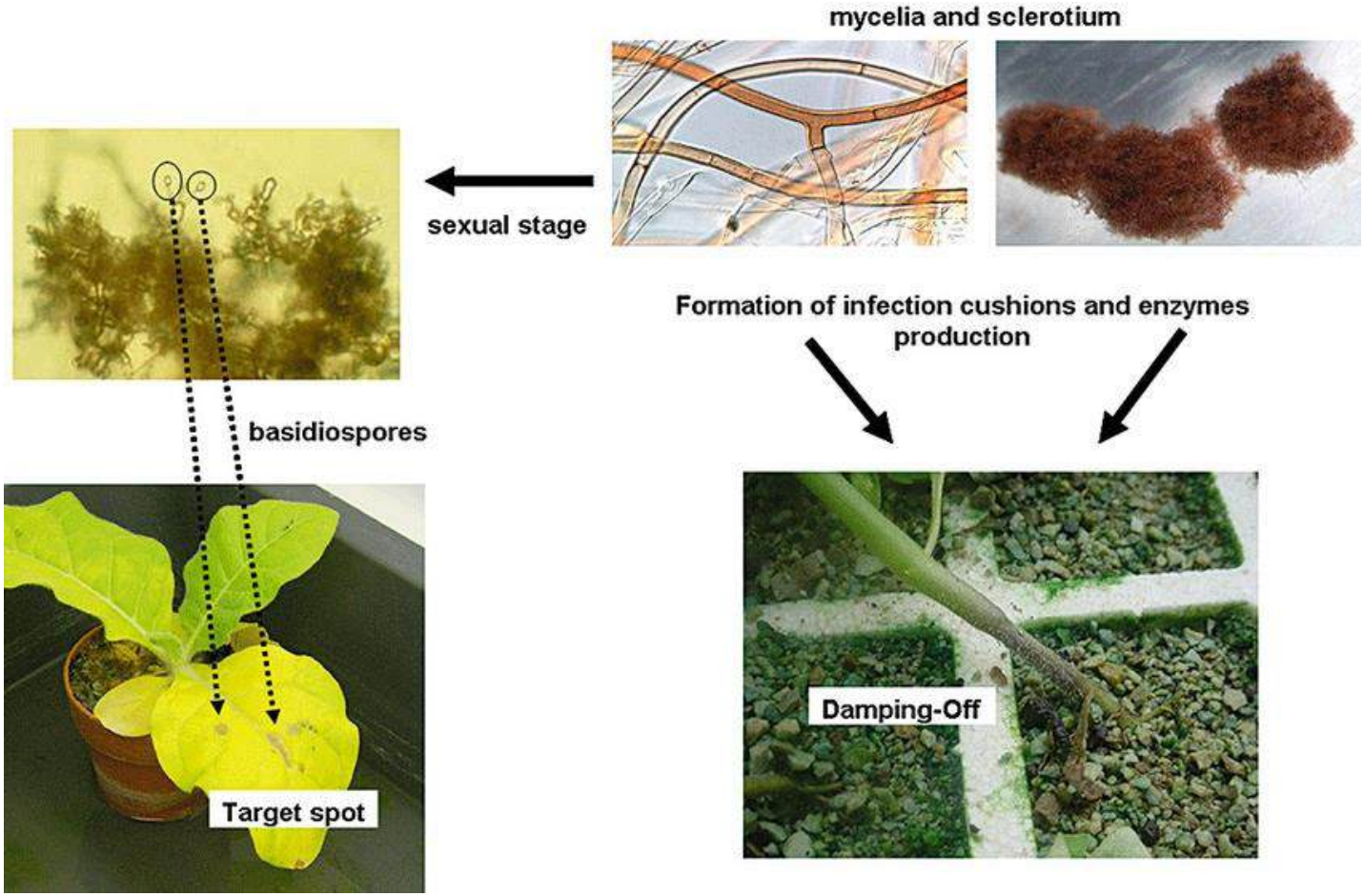
- The symptoms on leaves appear as small, circular, water soaked spots which later become tan to brown in colour and these spots coalesce to form larger areas
- The **pods are attacked at all stages of their growth** and dark brown, more or less circular, **slightly zonate and definitely sunken spots** which in moist weather are covered with hyphae and sclerotia of the fungi
- **Tan brown to reddish brown discoloration on infected seed**.

Root rot and web blight- *Rhizoctonia solani* (PS:*Thanatephorus cucumeris*)

Pathogen:

- The disease is caused by *Rhizoctonia solani*
- The **basidial stage of the fungus is *Thanatephorus cucumeris*** but occurs very rarely.
- The fungus consists of septate mycelium, the mycelium is branched near the distal septum of the mother hyphal cells **at right or acute angles** and the branches are **constricted at or near the point of origin of the septum**.
- In culture, sclerotia develop within 5-6 days.

Root rot and web blight- *Rhizoctonia solani*



Root rot and web blight- *Rhizoctonia solani*

Disease cycle and epidemiology:

- The pathogen perpetuates either through infected seed or in soil as sclerotia
- The **infected seeds after germination** may rot in the soil itself or killed by **post emergence damping off**
- The rain splashes spread the soil borne inoculum to foliage that leads to **web blight symptoms**.
- High rainfall and soil moisture with high relative humidity and soil temperature of 23 to 25°C favour disease development.

Root rot and web blight- *Rhizoctonia solani*

Management:

- Collection and destruction of infected plant debris.
- Follow long crop rotation.
- Soil solarization at least for 40-45 days during summer months.
- Soil amendment with neem or mustard cake (250g/m^2).
- Use healthy seed.
- Dry seed treatment with carbendazim (0.2%).

Root rot and web blight- *Rhizoctonia solani*

Management contd…

- Before the onset of monsoon rains**, apply pine needle/ grass mulch on the field floor** to avoid **splashing of pathogen inoculum to foliage.**

- Spray the crop with carbendazim (0.1%) or mancozeb (0.25%) or combination of mancozeb (0.25%) and carbendazim (0.05%) or tebuconazole (0.06%) and repeat at 10-14 days interval for foliar control of the disease.

Charcoal rot or dry root rot- *Macrophomina phaseolina*

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Charcoal rot or dry root rot- *Macrophomina phaseolina*

The disease is also known as **ashy stem blight, root rot, ashy grey stem** and **Macrophomina rot.**

Symptoms:

- The initial symptom is **yellowing of leaves, which soon dry up.**
- **The plant may wilt within** a week after the appearance of symptoms.
- When the stem is examined closely, **dark lesions may be seen on the bark** at the ground level.

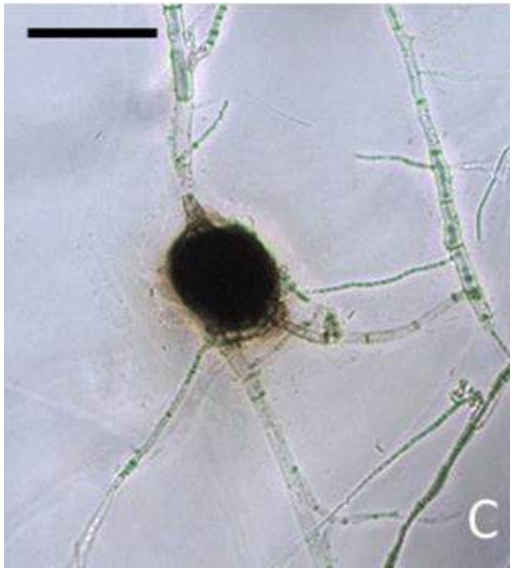
Charcoal rot or dry root rot- *Macrophomina phaseolina*

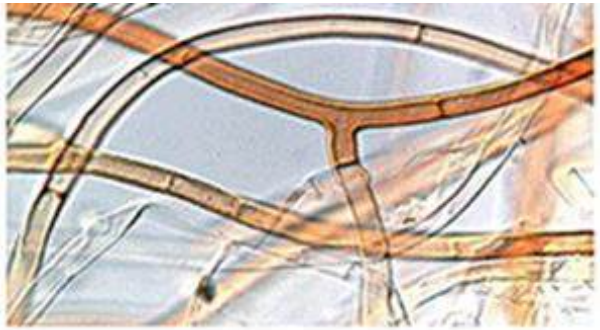
Symptoms contd….

- On uprooting infected plants, **dry rot symptoms on the basal stem and the main roots** may be seen.
- The **tissues are weakened and break off easily**.
- Black sclerotial bodies are formed on the diseased tissue a few days after infection.

Charcoal rot or dry root rot- *Macrophomina phaseolina*

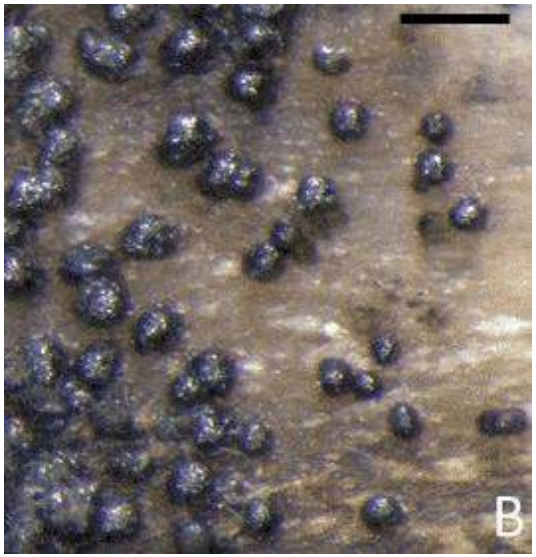
Sclerotial stage

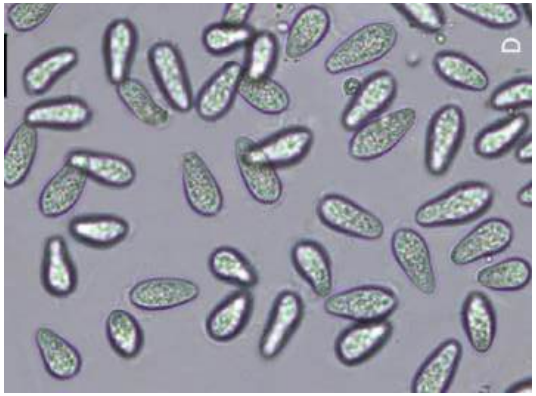




Pycnidial stage

Pycnidia





Pycnidiospores

Charcoal rot or dry root rot- *Macrophomina phaseolina*

Pathogen:

- The disease is caused by *Macrophomina phaseolina* (Pycnidial stage)
- The sclerotial stage of the fungus is known as *Rhizoctonia bataticola* (Taub.) Butler.
- The mycelium is **superficial or immersed, hyaline to brown, branched, septate and often tree-like in form**.
- The pycnidia are at first buried beneath the lead-grey epidermis in which many sclerotial bodies are generally mixed.

Charcoal rot or dry root rot- *Macrophomina phaseolina*

Pathogen contd..

- The **conidiophores are more or less straight**, sometimes crooked with a truncate tip.
- The conidia are one celled, more or less **fusiform, straight or slightly curved.**
- One end of the conidium is often pointed and the other is blunt.
- Sclerotia are readily produced, both on the plants in the field and in cultures.
- These are **jet black, smooth, hard round to oblong or irregular in shape**.

Charcoal rot or dry root rot- *Macrophomina phaseolina*

Disease cycle and epidemiology:

- The pathogen perpetuates as **sclerotia in the soil and in infected seed.**
- Survival is longer in dead host tissues than in free conditions in the soil.
- The fungus is **a poor competitor in soil but readily colonizes plant debris**.
- Population of the fungus increases when susceptible hosts are continuously grown in the field.

Charcoal rot or dry root rot- *Macrophomina phaseolina*

Disease cycle and epidemiology contd…

- When infected seed is sown in the field, it also introduces the fungus in the field.
- The sclerotia germinate and the mycelium cause primary infections.
- Environmental factors contributing to **soil moisture stress/ water deficit predispose** the plant to infection.
- **Soil temperature of 28°C** or more favours disease development.

Charcoal rot or dry root rot- *Macrophomina phaseolina*

Management:

- Collection and destruction of infected plant debris.
- Long crop rotations with non-leguminous crops.
- Solarization of infected soils during summer months can also help in reduction of soil borne inoculum.
- Use disease free seed and treatment with carbendazim or benomyl (0.2%).
- Application of biocontrol agents like *Trichoderma viride* and *Aspergillus niger* have also been found promising but need further confirmation